Proteomics assessment of CSF and blood plasma of the PPMI cohort using mass spectrometry-based untargeted proteome investigation

This document was prepared by Niveda Sundararaman and Rakhi Pandey in consultation with the **AMP PD Proteomics Working Group**.

AMP PD Proteomics Working Group Members: Amilcar Flores-Morales,Sanofi.com; Andy Christoforou; Bristol-Myers Squibb; Debra Babcock, National Institute of Neurological Disorders and Stroke; Bailin Zhang, Sanofi; Barry Landin ; Bradford Casey, The Michael J. Fox Foundation for Parkinson's Research; Sri Pullagura, Foundation for the National Institutes of Health (FNIH); Ethan.Xu, Sanofi; Howard Schulman, PPMI BRAC; Lyn Jakeman, National Institute of Neurological Disorders and Stroke; Marcus Bantscheff, GlaxoSmithKline; Pablo Sardi, Sanofi; Samantha Hutten, The Michael J. Fox Foundation for Parkinson's Research; Christine Swanson-Fischer, National Institute of Neurological Disorders and Stroke; Victoria Dardov, Technome

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Summary

Proteomics analysis was conducted on cerebrospinal fluid and blood plasma of both Parkinson's Disease patients and healthy participants in the PPMI cohort. Analysis was conducted using mass spectrometry on 2283 samples from 482 participants from cerebrospinal fluid and 949 samples from 179 participants from blood plasma. Mass spectrometry data was processed through the OpenSwath pipeline, run through mapDIA and batch corrected.

Abbreviations

CSF Cerebrospinal Fluid

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Associated Files

Batch Corrected Protein Expression Files
2023June20_PPMIProject196_Untargeted_PLASMA.csv
2023June20_PPMIProject196_Untargeted_CSF.csv
Sample Tissue Types: "CSF","Plasma"

File Structure

Column Name	**Column Description**
PROJECTID	PPMI project ID
SAMPLEID	PPMI sample ID
SMS_SUBJECT_ID	Subject ID
TESTNAME	Uniprot ID
TESTVALUE	Protein Abundance
UNITS	Units of protein abundance
RUNDATE	Day of mass spectrometry run

Method

Sample Preparation

Cerebrospinal fluid (CSF)

CSF Protein Digestion - Individual patient CSF samples and four CSF pooled samples from healthy individuals used as DCR were processed on a Beckman i7 automated liquid handling system in 96-well plate format using a method modified from our previous published methods for plasma and depleted plasma.[9-10] Briefly, the CSF sample were denatured and reduced by incubating with agitation at 60°C for 60 minutes with 30 µL of 55% 2,2,2-trifluoro-ethanol (TFE, Sigma), 14mM Dithiothreitol (DTT, Sigma) and dissolve in 40mM NH_4CO_3 (Ammonium bicarbonate, Sigma). Samples were alkylated by adding 10µL of 50mM iodoacetamide (IAA, Sigma) and incubated in the dark at room temperature for 30 minutes. The alkylation was quenched by adding 10uL of 100mM DTT to samples followed by a 15-minute incubation with agitation at room temperature. Sample solutions were then diluted with 180µL of 100mM

NH_4CO_3 to reduce TFE in solution concentration to ~6.5%. Samples were digested at 42°C for 4 hours in an Inheco incubator trypsin:protein ratio of 1:10. Sample digestion was quenched by adding 10 µL of 12.5% formic acid. The final peptide digest volume in the well was 265 µL, which was immediately sealed with foil adhesive seals and stored at -80°C until thawed for LC-MS analyses. Given the undigested CSF sample aliquot volumes of 25 µL and final peptide digest volumes of 265 µL as previously described, the following dilution scheme was used to target a peptide load of 500 ng for each sample injection.

Adding iRT Peptide Standards to CSF peptides- To prepare samples for LC-MS processing, sample peptides were thawed at room temperature and spun down, and 53µL were diluted into a second 96-well plate with 47µL 0.1% formic acid in water. 20µL of diluted peptides were then transferred to wells of a PCR plate containing 20µL of diluted commercial indexed retention time peptide standards (iRT) (Biognosys[WK1] ®). iRT standards were prepared by serial dilution, first diluting one vial of iRT peptides with 1000µL of 0.1% formic acid in water, then adding 90uL of the previous dilution to 910uL of 0.1% formic acid in water.

CSF Sample Desalting - CSF sample/iRT peptide mixtures were then desalted by loading 20µL of the mixture onto Evotips (EVOSEP, C18 disposable trap columns).

Plasma

Plasma Depletion - Plasma samples were depleted using 96 well plate previously described by McArdle and Binek *et al.* (2022), in which 14 most abundant proteins including albumin, immunoglobulins A, E, G, and M, kappa and lambda light chains, alpha-1-acidglycoprotein, alpha-1-antitrypsin, alpha-2- macroglobulin, apolipoprotein A1, fibrinogen, haptoglobin, and transferrin using the High Select Top 14 Abundant Protein Depletion Camel Antibody Resin (Thermo Fisher Scientific).[9]

The depletion protocol was previously described by McArdle and Binek et al., but here we scaled the method to allow for aliquoting depletion resin into 96-well plates in batches of 2 plates. Briefly, after equilibrating depletion resin at room temperature, the resin was poured into a 25mL pipetting reservoir affixed to an in-house made, 3D printed vortex attachment and shaken at 800 rpm to maintain a homogeneous resin slurry for consistent bead concentrations in each well of the 96-well plates containing diluted plasma samples (10 µL plasma diluted with 90 µL ammonium bicarbonate). Using a multichannel pipette, 300 µl of resin were added to all sample wells, and plates were sealed. Plates were then incubated in a plate shaker at 800 rpm for 1 hour at room temperature. Contents of the 96-well plates were then transferred to filter plates (Nunc) atop empty 96-well plates (Beckman Coulter), and the filter plate was sealed. The samples were then passed through the filter via centrifugation for 4 minutes at 100 rpm. Plates were then removed and rotated 180° then centrifuged for 4 minutes at 100 rpm. Resultant depleted proteins were then lyophilized via Speedvac and frozen at -80°C and stored until two more depleted plasma plates were ready for 4-plex tryptic digestion.

Naïve Tryptic Digestion and Desalting - Naïve plasma proteins were digested according to the protocol described by Fu *et al.* (2020)[2]. Naïve plasma tryptic peptide desalting was carried out using a positive pressure apparatus (Amplius Positive Pressure ALP, Beckman Coulter) mounted on the left side of the i7 workstation deck as described by Fu *et al.* (2020).[10]

Naïve plasma iRT Peptide Addition – Naïve plasma peptides were spiked 1:1 in MS vials with 1:20 diluted iRT peptides.

Depleted Plasma Digestion, iRT Addition and Desalting – Depleted plasma samples underwent the tryptic digestion procedure described by McArdle and Binek *et al.* (2022).[9] Resultant peptides were further diluted (5 µL peptides with 120 µL 0.1% formic acid in H_2O) then 25 uL of diluted peptides were spiked into 25uL of iRT standards (Biognosys®) that were diluted as described above for CSF Evotip loading. 20 uL of depleted plasma/iRT peptide mixtures were then desalted using Evotips.

LC-MS Methods

EVOSEP One LC System with Orbitrap Exploris 480 MS for CSF and Depleted Plasma

Data independent acquisition (DIA-MS) was implemented by loading CSF sample peptides from Evotips onto an EVOSEP EV1106 Analytical Column (EVOSEP, C18 AQ, 1.9µm beads, 150µm ID, 15cm long with an EVOSEP One LC system). Samples were injected into an Orbitrap Exploris 480 mass spectrometer (ThermoScientific) with an average flow rate of 1.5 µl/min using the 30SPD EVOSEP method. The MS settings used for data acquisition are as follows: for Global MS Settings, expected LC peak width was set to 20 seconds and default charge state set at 2^+. During the MS full scan data were collected in Profile form from 0 to 45 minutes over a scan range of 350-1400 m/z. Orbitrap resolution was set to 120k, while S-lens radio frequency was at 40%, normalized accumulated gain control (AGC) target was 300%, and maximum injection time was 45 milliseconds. DIA data were collected as Profile, scanning over a range of 200-1700 m/z with 50 isolation windows spanning 21 m/z with 1 m/z window overlap. Orbitrap resolution was 15k. Normalized collision energy was set to 28%, normalized AGC target was 100%, and maximum injection time was 22 milliseconds.

Depleted plasma tryptic peptides were separated by EVOSEP One LC System over a 45-minute gradient using the same column type as CSF and injected into an Orbitrap Exploris 480 MS using DIA-MS settings described by McArdle and Binek *et al.* (2022).[9]

Eksigent-415 LC System with Sciex Triple TOF 6600 MS for Naïve Plasma

Naïve plasma peptides were analyzed using the method described by Holewinski *et al.* (2016).[11] Briefly, samples data were acquired over a 60-minute gradient using a Eksigent-415 microflow

LC system coupled to a Sciex Triple TOF 6600. DIA-MS methods used 100 variable windows over a chromatographic gradient of 60 minutes in the 400–1200 m/z range.

Data Processing

These were converted to profile mzML format using MSConvert for .raw conversion and Proteowizard for .wiff conversion.

Intensity data for peptide fragments was extracted from mzML files using the open source openSWATH workflow[1] against the publicly available plasma library of the human twin population (February 2015) peptide assay library. Target and decoy peptides were then extracted, scored, and analyzed using the mProphet algorithm[2] to determine scoring cut-offs consistent with a 1% false discovery rate (FDR). Peak group extraction data from each DIA file was combined using the "feature alignment" script, which performs data alignment and modeling analysis across an experimental data set[3], and fragment-level data was normalized by MS2TIC.

Next, to obtain high-quality quantitative data, fragments from non-proteotypic peptides, i.e. peptides shared across different proteins[4], were discarded. The large data set showed irregular patterns of missingness distribution, invoking the need for data filtering to discard fragments having ≥50% missingness across all three biological groups (done using metadata provided from the partial unblinding).

Technical bias from batching during digestion was corrected using comBat corrective technique[5]. In the case of CSF samples, this was followed by Random Forest (RF) imputation[6]. In the case of plasma, the depleted and native batch corrected data was combined at the fragment-level following the dual plasma workflow as specified in Zhang et al[7].

The data was then processed using the mapDIA software[8] to roll up fragment-level data to peptide and protein levels. PPMI CSF batch corrected and imputed files are provided. PPMI Plasma batch corrected files are provided. .

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About the Authors

This document was prepared by **Niveda Sundararaman and Rakhi Pandey in Jennifer Van Eyk's group at Cedars Sinai Medical Center** in consultation with the **AMP PD Proteomics Working Group**.

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